

Planetary Motion

Python2 (?) code for basic earth-sun simulation

```
import ode, numpy as np
import visual as vp

def earth(id, r, v, t):
    global GM
    if id==0 : return v
    s = vp.mag(r) # magnitude
    return - GM*r/s**3

def go():
    r = np.array([1.017, 0.0])
    v = np.array([0.0, 6.179])
    scene = vp.display(
        title="Planetary motion",
        background=(0.2, 0.5, 1),
        forward =(0.2, -1),
        planet = vp.sphere(
            pos=r, radius=0.1,
            make_trail=True,
            material =vp.materials.earth,
            up=(0, 0, 1)
        ),
        sun = vp.sphere(
            pos=(0, 0),
            radius = 0.2,
            color=vp.color.yellow,
            material = vp.materials.emissive
        ),
        sunlight = vp.local_light(pos=(0, 0), color=vp.color.yellow)
    )
    t = 0.0
    h = 0.001
    while True:
        vp.rate(200)
        r, v = ode.leapfrog(earth, r, v, t, h)
        planet.pos = r
        if (scene.kb.keys):
            scene.kb.getkey(), scene.kb.getkey()

GM = 4* (np.pi**2)
go()
```

Terminology

- Aphelion : Position of Earth farthest from Sun
- Perihelion : Position of Earth closes to Sun
- 1 AU (astronomical usint) = 1.496×10^{11} m
- 1 year = 3.156×10^7 s

- 1 AU/year = 4740 m/s
- For the sun, $GM = 4\pi^2 \text{AU}^3 \text{year}^{-2}$

Reduction to one body problem

If we have bodies with masses m, M and positions $\vec{r}_1(t), \vec{r}_2(t)$ following gravity, then we can rewrite the system as :

$$\begin{aligned}\vec{R} &= \left(\frac{m}{m+M}\right)\vec{r}_1 + \left(\frac{M}{m+M}\right)\vec{r}_2 \quad [\text{CoM}] \\ \vec{r} &= \vec{r}_1 - \vec{r}_2 \\ m\ddot{\vec{r}}_1 &= \frac{GMm}{r^2}(-\hat{r}) \quad \text{and} \quad M\ddot{\vec{r}}_2 = \frac{GMm}{r^2}(\hat{r}) \\ \implies \ddot{\vec{R}} &= 0 \quad [\text{addition}] \\ \text{and } \ddot{\vec{r}} &= \frac{G(M+m)}{r^2}\hat{r}\end{aligned}$$

Just solve for $\vec{r}$ and you get $\vec{r}_1, \vec{r}_2$ easily. Since this is exactly the body orbiting-fixed-mass problem, we know the solutions are conic section.

Angular Momentum and Effective potential

Define $\omega = \dot{\theta}$ and $v_r = \dot{r}$ as angular and radial velocities. Note that v_r is different from $\vec{v} = \frac{d}{dt}\vec{r}$.

Define angular momentum as $\vec{L} = \vec{r} \times (m\vec{v})$.

We are assuming movement in x-y plane only, so $\vec{L} = L\hat{k}$.

Since the torque on the mass m (and the full system for reduced two body case) is 0, we have $\frac{d}{dt}\vec{L} = \vec{0}$.

This causes Kepler's second law (equal areas in equal times)

We want our numerical method to keep angular momentum constant too.

Another thing to keep constant is Energy (Hamiltonian), given by

$$E(\vec{r}, \vec{v}) = \frac{1}{2}mv^2 + V(r)$$

Here,

$$V(\vec{r}) = -\frac{GMm}{r}$$

is our usual gravitational potential.

Now, rewriting things in polar coordinates,

$$E(\theta, r, \omega, v_r) = \frac{1}{2}mv_r^2 + \frac{1}{2}m(r\omega)^2 + V(r)$$

But, due to constant L , we have :

$$mr^2\omega = L \implies r^2\omega^2 = \frac{L^2}{m^2r^2}$$

which is a function of $\vec{r}$, and not ω .

Thus, we have :

$$E(r, v_r) = \frac{1}{2}mv_r^2 + V_{\text{eff}}(r)$$

$$\text{where } V_{\text{eff}}(r) = \frac{1}{2} \frac{L^2}{mr^2} - \frac{GMm}{r^2}$$

The $\frac{L^2}{2mr^2}$ term is the "centrifugal potential energy". This arises because we have effectively shifted to a rotating reference frame.

The equation is now entirely one-dimensional (since $v_r = \dot{r}$).

Usually, for bounded motion (elliptical trajectories), the values r_{max} and r_{min} are called "turning points", which is also where the energy reaches a maximum or minimum value.

Kepler Orbits

In polar coordinates, the solution to the two body problem is :

$$r(\theta) = \frac{L^2}{mk} \left[\frac{1}{1 + e \cos(\theta + \theta_0)} \right]$$

where $e = \sqrt{1 + \frac{2EL^2}{mk^2}}$ [eccentricity]

Here $k = GMm$.

The shape of the orbit is determined as :

Energy	eccentricity	shape
+ve	> 1	hyperbola
0	1	parabola
-ve	< 1	ellipse
$-\frac{mk^2}{2L^2}$	0	circle

For elliptical orbits, we have :

$$a = \frac{-k}{2E}$$

$$b = a\sqrt{1 - e^2} = \frac{-k}{2E} \sqrt{\frac{-2EL^2}{mk^2}} = \sqrt{-\frac{L^2}{2mE}} = \frac{L}{\sqrt{-2mE}}$$

$$r(\theta) = \frac{a(1 - e^2)}{1 - e \cos(\theta + \theta_0)}$$

Setting $\theta_0 = \pi$ (body starts at perihelion), we get :

$$r(\theta) = \frac{a(1 - e^2)}{1 - e \cos \theta}$$

The tangential velocity components at aphelion (r_{max}) and perihelion (r_{min}) are :

$$v_{\text{ap}} = \sqrt{\frac{k}{ma} \left[\frac{1 - e}{1 + e} \right]}$$

$$v_{\text{peri}} = \sqrt{\frac{k}{ma} \left[\frac{1 + e}{1 - e} \right]}$$

Since the angular momentum is fixed, the area is swepted at a rate of $\frac{L}{2m} = b\sqrt{-2mE}/(2m) = \frac{b\sqrt{-E}}{\sqrt{2m}} = b\sqrt{\frac{k}{4am}}$.

The total area is πab . Thus, the time period is $T = \pi a \sqrt{\frac{4am}{k}} = \sqrt{4\pi^2 a^3 \frac{1}{GM}}$. This gives Kepler's 3rd law :

$$T^2 = \frac{4\pi^2 a^3}{GM}$$

Using this for the sun-earth system, we have :

$$(1 \text{ year})^2 = 4\pi^2 (\text{AU})^3 \frac{1}{GM_s}$$

Thus, $GM_s = 4\pi^2$ using AU and years.

Stable orbits

If by changing the energy by a small amount, the orbits doesn't change qualitatively (is still closed), it's a stable orbit.

A closed orbit has the ratio of number of oscilations in r (namely $n_r(t)$) and oscilations in θ (namely $n_\theta(t)$) to be a constant ($n_r/n_\theta = \text{constant}$) .

Only forces with a $1/r^2$ or r proportionality can produce closed orbits. Other kind of forces/potentials cause orbital precession.

Precesion of platenary orbits

While classic gravity *does* follow the inverse square law, the planets in the solar system still undergo precession. This has two reasons.

The first is that other planets also tug and pull at a given planet. This can be simulated numerically simply using newton's theory of gravity.

For example, doing this for mercury give a precession of 574 arc-seconds per century (1 arc-second is 1/3600 of a degree), or 0.16 degrees per century. This rate is $\frac{d\theta_0}{dt}$ where θ_0 is the aphelion.

Even with this, there is a leftover precession of 43 arc-seconds, which is due to the second reason.

The second reason is that, according to general relativity, gravity isn't entirely an inverse-square force. Instead, we have :

$$F = \frac{GMm}{r^2} + \lambda \frac{GMm}{r^4}$$

Rung-Lenz vector

We define it as

$$\vec{A} = \vec{p} \times \vec{L} - mk\hat{r}$$

This is 0 for circular orbits and is a constant vector pointing to the perihelion (negative x axis direction) in case of closed elliptical orbits.

The rate of change of the angle that $\vec{A}$ makes with x-axis is the rate of precession.

Leapfrog with time transformation

We scale the time step h based on the distance r with a scaling factor $\Omega(r)$. Basically, rather than using t as the path parameter, we use $s(t)$ with the property that $\frac{ds}{dt} = \Omega(r)$ is large when r is small. So now, we need to solve for $\vec{v}(s), \vec{r}(s), t(s)$ rather than $\vec{v}(t), \vec{r}(t)$.

This means :

$$\frac{d}{ds}\vec{r} = \vec{v}/\Omega(r)$$

$$\frac{d}{ds} \vec{v} = \vec{a}(\vec{r})/\Omega(r)$$

$$\frac{d}{ds} t = 1/\Omega(r)$$

In position-verlet Leapfrog, the way a single step works is through

$$\vec{z}_{n+1} = \exp(D_T h/2) \exp(D_V h) \exp(D_T h/2) \vec{z}_n$$

and thus we require the time step h to be constant.

Having time step $\Omega(r_n)h$ in the first sub-step and $\Omega(r_{n+1/2})h$ in other two will make it wrong.

So, we use the same step scale, call it W , which is computed at the very start. So our equations become :

$$\frac{d}{ds} \vec{r} = \vec{v}/W$$

$$\frac{d}{ds} t = 1/W$$

$$\frac{d}{ds} \vec{v} = \vec{a}(\vec{r})/\Omega(r)$$

$$\frac{d}{ds} W = \frac{d}{ds} \Omega(r) = \nabla \Omega(r) \cdot \frac{d\vec{r}}{ds} = \nabla \Omega(r) \cdot \vec{v}/\Omega(r)$$

The reason for still writing $\Omega(r)$ is so that this system can be directly put into a leapfrog solver function by thinking of $\vec{q} = (\vec{r}, t)$ as position and $\vec{p} = (\vec{v}, W)$ as velocity vector.

This is not actually necessary (doubt) and doesn't even make physical sense since the $\vec{q}, \vec{p}$ don't correspond to a real $T(\vec{p}), V(\vec{q})$ pair. We could just compute $W = \Omega(r)$ directly before doing the leap-frog step and update t separately. This is a more sane approach IMO.

Their choice of $\Omega(r)$ is $1/r$.

In their update equations, they don't actually have the same scaling factor for the substeps. I really don't understand what's going on at this point.

The updates are :

$$\vec{r}_{1/2} = \vec{r}_0 + \vec{v}_0 \frac{h}{2} \frac{1}{W_0}$$

$$t_{1/2} = t_0 + \frac{h}{2} \frac{1}{W_0}$$

$$\vec{v}_1 = \vec{a}(\vec{r}_{1/2}) r_{1/2} h$$

$$W_1 = W_0 - \vec{r}_{1/2} \cdot \left(\frac{\vec{v}_0 + \vec{v}_1}{2} \right) h \frac{1}{r_{1/2}^2}$$

$$\vec{r}_1 = \vec{r}_{1/2} + \vec{v}_1 \frac{h}{2} \frac{1}{W_1}$$

$$t_1 = t_{1/2} + \frac{h}{2} \frac{1}{W_1}$$

Other planets' effect

To get the net effect of other planet's gravitational field over centuries, we can do a central field approximation. Essentially, over centuries, the other planet has been at all positions of a (hypothetical) ring with mass distributed over regions in proportion to the time of a "year" that the planet spends in that region.

We can do something even simpler and just consider a circular orbit, making the ring uniform.

Then, the non-relativistic gravitational potential for such a planet with orbit radius a when mercury is at distance $r = \rho a$ from sun, will be :

$$V(r) = -\frac{Gm}{a} \frac{M}{2\pi a} \int_0^{2\pi} \frac{d\theta}{\sqrt{1 + \rho^2 - 2\rho \cos \theta}}$$

Since $\rho < 1$ for mercury and any other planet, we can (somehow) write this as a power series :

$$V(r) = -\frac{GM_p m}{a} \left[1 + \frac{1}{4}\rho^2 + \frac{9}{64}\rho^4 + \frac{25}{256}\rho^6 + O(\rho^8) \right]$$

Then, we take the negative gradient to get the force and then the acceleration :

$$\vec{a}_p = \frac{GM_p}{2a^3} \vec{r} \left[1 + \frac{9}{8}\rho^2 + \frac{75}{64}\rho^4 + O(\rho^6) \right]$$

Note the sign. It's away from the sun.

Relation of Rung-Lenz vector with $SO(4)$.

We already know that for any scalar quantity $f(\vec{q}, \vec{p})$, we have :

$$\frac{d}{dt} f = \{f, H\}$$

This is Liouville's theorem. This then carries over easily to statistical mechanics, but for the probability density ρ .

There's an analogous theorem in Quantum Statistical Mechanics for density matrices ($\hat{\rho} = |\psi\rangle \langle\psi|$) which is very useful when you have mixed states. The analogous theorem I talked about is :

$$i\hbar \frac{\partial}{\partial t} \hat{\rho} = [\hat{H}, \hat{\rho}] = \hat{H}\hat{\rho} - \hat{\rho}\hat{H}$$

It seems all of physics is littered with commutator, for the uncertainty principle to classical mechanics.

Now, as with all quantities, the Rung-Lenz vector's components also follows Liouville's theorem, giving us:

$$\frac{d}{dt} \vec{A} = \{\vec{A}, H\}$$

Using the definition of $\vec{A}$, we can write :

$$\{\vec{A}, H\} = \{\vec{p} \times \vec{L}, H\} - GMm\{\hat{r}, H\}$$

But then using product rule for terms of form $p_i L_j$, we can easily get :

$$\begin{aligned} \{\vec{p} \times \vec{L}, H\} &= \{\vec{p}, H\} \times \vec{L} + \vec{p} \times \{\vec{L}, H\} \\ &= \left(\frac{d}{dt} \vec{p} \right) \times \vec{L} + \vec{p} \times \left(\frac{d}{dt} \vec{L} \right) \end{aligned}$$

Now, since the force is radial, we have $\frac{d}{dt} \vec{L} = 0$. As for $\frac{d}{dt} \vec{p}$, that's just the force, namely $\vec{F} = -\frac{GMm}{r^2} \hat{r}$. So, this leaves us with :

$$\{\vec{p} \times \vec{L}, H\} = -\frac{GMm}{r^2} \hat{r} \times \vec{L}$$

But remember, $\vec{L} = \vec{r} \times \vec{p}$, allowing us to write :

$$\begin{aligned} \{\vec{p} \times \vec{L}, H\} &= \frac{GMm}{r} ((\hat{r} \cdot \hat{r}) \vec{p} - (\hat{r} \cdot \vec{p}) \hat{r}) \\ &= \frac{GM}{r} (\vec{p} - (\hat{r} \cdot \vec{p}) \hat{r}) \end{aligned}$$

Now, for the other term, we have :

$$\begin{aligned}
GMm \frac{d}{dt} \hat{r} &= GMm \frac{d}{dt} \frac{\vec{r}}{r} = GMm \left(-\frac{1}{r^2} \frac{dr}{dt} \vec{r} + \frac{1}{r} \vec{v} \right) \\
&= \frac{GMm}{r} \left(-\frac{1}{r} v_r \vec{r} + \vec{v} \right) = \frac{GMm}{r} \left(-(\vec{v} \cdot \hat{r}) \hat{r} + \vec{v} \right) \\
&= \frac{GM}{r} \left(-(\vec{p} \cdot \hat{r}) \hat{r} + \vec{p} \right)
\end{aligned}$$

Thus, subtracting these two will give 0. Hence we have proved that $\{\vec{A}, H\} = 0$, and thus $\frac{d}{dt} \vec{A} = 0$.

We already know that $\frac{d}{dt} \vec{L} = 0$. What this means is that any function of the *components* of $\vec{A}$ and $\vec{L}$ is also conserved (due to multivariable chain rule).

In fact, due to the antisymmetry property of poisson bracket, we have

$$\frac{d}{dt} \{a, b\} = \{\{a, b\}, H\} = \{a, \{b, H\}\} - \{b, \{a, H\}\}$$

Thus, any poisson bracket $\{L_i, A_j\}, \{L_i, L_j\}, \{A_i, A_j\}$ is also a constant across time.

Define $\vec{D} = \vec{A}/\sqrt{-2mE}$

The poisson brackets of all elements with all other elements is give as follows :

	D_1	D_2	D_3	L_1	L_2	L_3
D_1	0	L_3	$-L_2$	0	D_3	$-D_2$
D_2	$-L_3$	0	L_1	$-D_3$	0	D_1
D_3	L_2	$-L_1$	0	D_2	$-D_1$	0
L_1	0	D_3	$-D_2$	0	L_3	$-L_2$
L_2	$-D_3$	0	D_1	$-L_3$	0	L_1
L_3	D_2	$-D_1$	0	L_2	$-L_1$	0

Now, for the $so(4)$ group of antisymmetric 4x4 rotation matrices J_{ij} that rotate in the ij plane, we can define the commutator $[a, b] = ab - ba$ as the group operation. This then gives us the next *commutator* table :

	J_{12}	J_{13}	J_{14}	J_{23}	J_{24}	J_{34}
J_{12}	0	$-J_{23}$	$-J_{24}$	J_{13}	J_{14}	0
J_{13}	J_{23}	0	$-J_{34}$	$-J_{12}$	0	J_{14}
J_{14}	J_{24}	J_{34}	0	0	$-J_{12}$	$-J_{13}$
J_{23}	$-J_{13}$	J_{12}	0	0	$-J_{34}$	J_{24}
J_{24}	$-J_{14}$	0	J_{12}	J_{34}	0	$-J_{23}$
J_{34}	0	$-J_{14}$	J_{13}	$-J_{24}$	J_{23}	0

Turns out that the components of $\vec{L}, \vec{D}$ are the generators of a *Lie algebra* that is isomorphic to $so(4)$. A Lie Algebra has two operations; addition, which is some sort of abelian/commutative operation that the elements (including the identity "0") form a group over; and the Lie Bracket which is a binary operation $[\cdot, \cdot] : G^2 \rightarrow G$ need not have a group.

The Lie Algebra in question is of course $so(4)$.

Star Wobbles and Exoplanets

RV is "Radial Velocity" method. This is used to detect wobbling of a star. It uses the Doppler effect (for light) to get the (radial) velocity. (Note : "radial" means the direction from the observer at the origin to the star).

Remember that for a planet with position $\vec{r}_1$ and star with position $\vec{r}_2$, and masses m, M respectively, we had :

$$\begin{aligned}\vec{R} &= \frac{m\vec{r}_1 + M\vec{r}_2}{m + M} \\ \vec{r} &= \vec{r}_1 - \vec{r}_2 \\ \Rightarrow \left(\frac{m + M}{m}\vec{R} - \vec{r}\right) &= \left(\frac{M}{m} + 1\right)\vec{r}_2 \\ \Rightarrow \left(\frac{m + M}{m}\right)\vec{v}_C - \vec{v} &= \left(\frac{M}{m} + 1\right)\vec{v}_2\end{aligned}$$

Here, $\vec{v}_2 = \frac{d}{dt}\vec{r}_2$ is the velocity of the star and $\vec{v} = \frac{d}{dt}\vec{r}$ is the relative velocity of the planet. One can define $\vec{v}_* = \vec{v}_2 - \vec{R}$ as the relative velocity of the star wrt the center of mass, giving us :

$$\vec{v}_* = -\left(\frac{m}{M + m}\right)\vec{v}$$

Now, $\vec{r}$ changes as per the acceleration $\vec{a}(\vec{r}) = -\hat{r}G(M + m)/r^2$ and can be computed numerically using the code we have written. This will then give us $\vec{v}$ and then $\vec{v}_*$.

Then, the component of $\vec{v}_*$ in the direction of (or away from) the observer (not to be confused with $v_r = \vec{v} \cdot \hat{r}$).

Now, the orbit isn't necessarily in the "x-y" plane, but at an inclination.

To make things simple, let's define $\hat{z}$ as the direction that the observer is looking. Then, let $\hat{n}$ be a vector normal to the actual plane in which the star and planet are. Then, the inclination i of $\hat{n}$ with $\hat{z}$ can be estimated using Doppler effect.

Let's also assume the rotation done is about $\hat{x}$ as the axis. Thus, the x-axis is the line joining the points of intersection of the x-y plane and the orbit of the planet.

Let θ_a (called ω in the chapter 4 of the book) be the angle that the aphelion $\vec{r}_a$ makes with $\hat{x}$.

It's easier to think of the orbit as being transformed from the usual setup by the rotation $R_x(i)R_z(\theta_a)$.

With $\hat{z}$ as the line of sight, we have v_{*z} as the "radial velocity".

Define the distance from the focus for a standard ellipse as :

$$r_s(\theta) = a(1 - e^2)/(1 - e \cos \theta)$$

Here, θ is the angle that the ray joining the focus to the point makes with the x-axis.

Then, the distance from focus at angle θ from x-axis for an ellipse rotated by θ_a about the focus is $r_s(\theta - \theta_a)$. The y coordinate is then $y = r_s(\theta - \theta_a) \sin \theta$. The velocity component along $\hat{y}$ is then:

$$\begin{aligned}\frac{d}{dt}y &= \omega[r'_s(\theta - \theta_a) \sin \theta + r_s(\theta - \theta_a) \cos \theta] = \frac{2\pi ab}{T} \left[\frac{r'_s}{r_s^2}(\theta - \theta_a) \sin \theta + \frac{1}{r_s} \cos \theta \right] \\ &= \frac{2\pi ab}{T(1 - e^2)} \left[-\frac{d}{d\theta}(1/r_s)(\theta - \theta_a) \sin \theta + (1/r_s)(\theta - \theta_a) \cos \theta \right] \\ &= \frac{2\pi b}{T(1 - e^2)} \left[-e \sin(\theta - \theta_a) \sin \theta + (1 - e \cos(\theta - \theta_a)) \cos \theta \right] \\ &= \frac{2\pi a}{T\sqrt{1 - e^2}} [\cos \theta - e \cos \theta_a]\end{aligned}$$

This is without the rotation $R_x(i)$ about the x-axis and only with the rotation $R_z(\theta_a)$.

On applying the rotation, the x component of the velocity doesn't change, but the y component and z component are now

$$v_y = \frac{2\pi a}{T\sqrt{1-e^2}}(\cos\theta - e\cos\theta_a)\cos i$$

$$v_z = \frac{2\pi a}{T\sqrt{1-e^2}}(\cos\theta - e\cos\theta_a)\sin i$$

Thus, the radial velocity is :

$$v_z = \frac{2\pi a}{T\sqrt{1-e^2}}\sin i(\cos\theta - e\cos\theta_a)$$

This then gives the radial velocity for the star as :

$$v_{*z} = \gamma(\cos\theta - e\cos\theta_a)$$

$$\text{where } \gamma = \frac{m}{m+M} \frac{2\pi a}{T\sqrt{1-e^2}} \sin i$$

Note that in the book (and in the lab sheet) we defined θ using ray from focus to aphelion as reference ray. But here, we are using the x-axis as the referce.

Going back to the convention used in the book, we should write $v_{*z} = \gamma(\cos(\theta + \theta_a) - e\cos\theta_a)$.

All of this was done in the CoM frame of reference. Moving to the observer frame, we have the actual radial velocity $C_z + v_{*z} = C_z + \gamma(\cos(\theta + \theta_a) - e\cos\theta_a)$ where $\vec{C} = \frac{d}{dt}\vec{R}$.

Although simulating θ as a function of t for a given e, a, T is possible, the best method is to use Kepler's equation and vary the "eccentricity anamoly" ψ to produce :

$$t(\psi) = \frac{T}{2\pi}(\psi + e\sin\psi)$$

$$\cos\theta(\psi) = \frac{e + \cos\psi}{1 + e\cos\psi}$$

Otherwise, we can compute the speed and distance at aphelion as :

$$v_a = \frac{2\pi a}{T} \sqrt{\frac{1-e}{1+e}}$$

$$r_a = a(1+e)$$

Then, we just need to set these as the initial conditions for our numerical integrator.

But using the Kepler's equation is exact; so we will do that.

Book

- "Computational Physics" by Giordano, Naganishi
- Show that Rung Lenz is constant both computationally and analytically.
- Show that Rung Lenz vector is related to $SO(4)$